CLASSIFIED: SOVEREIGN DEFENSE PROTOTYPE // 100% AIR-GAPPED

KLS HACKFEST 2026 // PROBLEM STATEMENT 1A

PROJECT RAKSHAK 2.0

Sovereign Dual-Domain C4ISR & Multi-Modal Geospatial Threat Intelligence Platform

ENGINEERING UNIT: TEAM BOTS (KLS GIT / KLE TECH)

TARGET ENVELOPE: EMCON-ALPHA (ZERO CLOUD EGRESS)

BENCHMARK BASIS: VERIFIABLE RAW JSON AUDIT

TEAM MEMBERS // TEAM BOTS

ADITYA BAJANTRI LEAD / VISION	GAGAN BONGALE DATA TILING
UTSAV NANAPUR RADAR / AIS	MISBAH FALAK KINEMATICS / THREAT
SHALINA MANIYAR C2 INTERFACE	NIYATI GOGRI STANAG RAG

Executive Note for Technical Judges: Project Rakshak 2.0 addresses high-threat littoral and northern border operations where forces operate under electronic warfare radio silence (EMCON) and denied communications (DDIL). Every architectural module below was individually engineered and benchmarked by the designated team member. All figures reflect raw, verifiable evaluation reports stored in `evaluation/results/`.

PILLAR I: TACTICAL VISION & GEOSPATIAL SLICING ENGINEERING

ADITYA BAJANTRI — LEAD / VISION SYSTEMS

SYSTEM ARCHITECT & COMPUTER VISION

- **System Architecture & Sovereign Air-Gap:** Directed the 4-tier C4ISR platform architecture, enforcing strict zero-egress boundaries (0 non-loopback sockets, 0 external DNS queries, verified via `egress_test_report.json`).
- **Dual-Model Satellite Computer Vision:** Trained and fine-tuned dual optical detection models (YOLO11m 20.1M generalist + YOLO11n 2.6M specialist) on 0.3m GSD sub-meter satellite imagery (xView), achieving **48.46% overall mAP@50** and **89.85% mAP@50 on aircraft**.
- **Ensemble & Edge Acceleration:** Engineered Weighted Box Fusion (WBF) and FP16 CUDA/TensorRT acceleration, clocking **17.85 ms inference latency (~56 FPS)** on host tactical GPU hardware.
- **Multi-Temporal Satellite Change Detection:** Built the bi-temporal change detection pipeline combining ORB feature extraction and RANSAC sub-pixel co-registration (**<0.15 px registration error**) with Hungarian bipartite matching to autonomously detect added, removed, and relocated military assets.

`Primary Model: YOLO11m (20.1M)` `Aircraft mAP@50: 89.85%` `Boosted Latency: 17.85 ± 0.11 ms` `Co-Registration: 0.1449 px Error` `Network Egress: 0 Bytes (Air-Gapped)`

GAGAN BONGALE — DATA TILING & GEOSPATIAL PIPELINE

LARGE-SCALE GEOSPATIAL INGESTION

- **Sliced Geospatial Inference (SAHI):** Engineered the sliced tiling engine to ingest and process multi-gigabyte satellite scenes and large-format GeoTIFFs without memory overflow or compute bottlenecks.
- **Dynamic 1024×1024 Overlapping Slicer:** Engineered the sliding-window slicing algorithm with overlap compensation, specifically calibrated to preserve small, low-pixel-footprint tactical targets (12–35 px maritime craft).
- **Coordinate Transformation & Georeferencing:** Converted raw pixel bounding boxes to real-world UTM and WGS84 geographic coordinates, achieving sub-meter ground localisation accuracy (**CEP50 of 0.70 m**, **CEP90 of 2.43 m**).
- **Air-Gapped Procedural Basemaps & Data Balancing:** Developed local offline vector tile generation (`tile_service.py`) eliminating external Mapbox/OSM calls, along with class-weighted sampling algorithms to resolve severe training set imbalances.

`Tiling Window: 1024×1024 Overlapping` `Localisation Accuracy: 0.70 m CEP50` `CEP90 Precision: 2.43 m` `Scene Ingestion: 1.62 s / Scene` `Basemaps: 100% Procedural Offline`

PILLAR II: RADAR / AIS CROSS-CORRELATION & KINEMATIC THREAT TRACKING

UTSAV NANAPUR RADAR / AIS STREAM FUSION

SPACEBORNE SAR & MARITIME INTELLIGENCE

- **Sentinel-1 SAR Radar Processing:** Developed the spaceborne Synthetic Aperture Radar (SAR C-Band) detection engine using a 2D CA-CFAR (Cell-Averaging Constant False Alarm Rate) algorithm that screens 25,000 radar cells in **1.1 seconds**.
- **Kinematic AIS Dead-Reckoning:** Ingested live maritime AIS transponder feeds and implemented Dead-Reckoning (SOG/COG) temporal projection to maintain accurate target correlation across ±5 to ±30-minute sensor time offsets.
- **Dark Vessel Interception:** Automated Haversine great-circle spatial cross-matching (**85 ms latency**) between SAR radar contacts and AIS positions, achieving a **79.40% headline capture rate** of non-broadcasting / dark vessels with **100% precision**.
- **Naval Hull Classification:** Implemented physical waterline length estimation to automatically categorize radar contacts into defense naval classes (Skiffs, Patrol Trawlers, Corvettes, Frigates, and Capital Ships).

`SAR Algorithm: 2D CA-CFAR` `Processing Time: 1.1 s (25k Cells)` `Matching Latency: 85 ms` `Dark Capture Rate: 79.40 ± 2.80%` `Alert Precision: 100.0% (0 False Darks)`

MISBAH FALAK KINEMATICS / THREAT & AUTO-TRIAGE

MULTI-TARGET TRACKING & THREAT SCORING

- **Multi-Sensor Kinematic Tracking:** Implemented a Constant-Velocity Kalman Filter in a local ENU frame fusing multi-sensor covariances (AIS σ=12m, Optical σ=25m, SAR σ=65m).
- **Track Association & Zero ID-Switches:** Implemented Joint Probabilistic Data Association (JPDA) and Hungarian bipartite gating, maintaining **0 track ID switches** through 60-second AIS blackouts and sharp 90° tactical evasion maneuvers.
- **Deterministic Tactical Threat Matrix:** Formulated the real-time threat evaluation engine (0–100 scoring) calculating kinematic vectors, radar cross-sections, transponder status, and proximity to sensitive military perimeters (e.g., Mumbai ODA, Border Sector Alpha).
- **Analyst Cognitive Workload Reduction:** Designed 500m spatial entity clustering and rule-based auto-triage, delivering an **86.54% reduction in items requiring manual human review** with **0.0% missed critical threats**.

`Tracking Filter: CV Kalman + JPDA` `ID Switches: 0 (Evasion Turns)` `Item Reduction: 86.54% (14k → 1.8k)` `Auto-Close Benign: 20.72%` `Missed Threats: 0.0% (Safety Suite v2)`

Cross-Sensor Kinematic Fusion Architecture: Radar hits extracted by Utsav's CA-CFAR engine are cross-correlated against dead-reckoned AIS transponders. Contacts exhibiting spatial dissonance (>2.0 km discrepancy) or completely absent transponder signatures are flagged as *Dark Vessels*. Misbah's tracking engine ingests these entities, assigns a persistent Kalman state vector, and runs dynamic threat-scoring against defined exclusion perimeters (e.g., offshore oil platforms or border defense zones).

PILLAR III: TACTICAL C2 VISUALIZATION & AUTONOMOUS MILITARY DOCTRINE

SHALINA MANIYAR C2 INTERFACE & COMMON OPERATING PICTURE

TACTICAL GIS & COMMAND DASHBOARD

- **Tactical Command & Control (C2) Frontend:** Designed and built the military Common Operating Picture (COP) using React, Vite, and TypeScript for real-time situational awareness.
- **Zero-Egress Vector GIS Engine:** Engineered an air-gapped SVG/Canvas geospatial rendering system (`world_land.json`, `offlineGeoData.ts`), completely removing external tile server calls to prevent RF and DNS signature leakage.
- **Role-Based Operational Workspaces:** Delivered tailored views for **Tactical Analysts** (split-screen triage queues, one-click dark vessel adjudication, target fly-to inspection) and **Commanders** (tactical heatmaps, zone breach alerts, and intercept dispatching).
- **Mission Dispatch & Telemetry Display:** Implemented real-time sensor-to-screen telemetry streaming, threat priority queuing, and interactive waypoint routing for field units and patrol craft.

`Tech Stack: React 18 / Vite / TypeScript` `Map Engine: 100% Offline SVG/Canvas` `Triage Queues: Priority vs Review` `External Calls: 0 (No OSM/Mapbox)`

NIYATI GOGRI STANAG SITREP & SOVEREIGN DOCTRINE RAG

DEFENSE INTELLIGENCE & OODA OPTIMIZATION

- **Air-Gapped Defense Doctrine RAG:** Built an offline BM25 lexical retrieval engine indexing sovereign military doctrines (Indian Navy Rules of Engagement INBR 8, UNCLOS maritime law, Army Border SOPs) with **<40 ms query response time** and zero external cloud LLM dependencies.
- **Automated NATO STANAG 2014 SITREP Compiler:** Implemented the tactical Situation Report compiler, procedurally converting fused multi-domain telemetry (optical, SAR, UGS seismic, AIS) into standardized defense intelligence briefs in **<50 ms (16.7 ms REST runtime)**.
- **OODA Decision Cycle Compression:** Slashed operational mission-planning cycle time by **96.5%**—compressing a traditional 24-minute manual ops-room procedure down to under **50 seconds operator-paced** (<0.2s machine execution).
- **Resilient DDIL Synchronization:** Integrated a store-and-forward SQLite WAL replication pipeline, ensuring **100% alert delivery** and zero packet loss over jammed, degraded, or intermittent 32 kbps tactical combat radio links.

`Doctrine Index: Local BM25 Lexical` `RoE Query Latency: < 40 ms` `SITREP Standard: STANAG 2014 (<50 ms)` `OODA Reduction: 96.51% (24m → 50s)` `DDIL Deliverability: 100.0% (SQLite WAL)`

SYSTEM INTEGRATION MATRIX: THE C4ISR OODA LOOP KILL-CHAIN

OODA STAGE	OPERATIONAL PILLAR	LEAD ENGINEER	CORE SUBSYSTEM	AUDITED BENCHMARK
1. OBSERVE	Geospatial Data Tiling	**Gagan Bongale**	SAHI Dynamic 1024px Tiling	`1.62 s / Scene, 0.70m CEP50`
1. OBSERVE	Optical Satellite AI	**Aditya Bajantri**	YOLO11m + YOLO11n WBF	`17.85 ms, 48.46% mAP@50`
2. ORIENT	SAR & AIS Fusion	**Utsav Nanapur**	Sentinel-1 2D CA-CFAR + SOG/COG	`79.40% Dark Vessel Capture`
2. ORIENT	Kinematic Threat Triage	**Misbah Falak**	Kalman + JPDA + Threat Matrix	`0 ID Switches, 86.54% Workload Cut`
3. DECIDE	Doctrine Advisory & SITREP	**Niyati Gogri**	Sovereign BM25 RAG + STANAG	`<40 ms RoE, <50 ms SITREP`
4. ACT	Tactical C2 Interface	**Shalina Maniyar**	Zero-Egress React C2 Dashboard	`0 Outbound Calls, Real-time Sync`

Judges Briefing Conclusion: *"Every member of Team BotS owned an indispensable operational phase of modern digital warfare: from raw orbital sensor ingestion (Gagan) and multi-modal edge AI (Aditya, Utsav), through mathematical kinematics and cognitive auto-triage (Misbah), to sovereign doctrine synthesis (Niyati) and tactical command visualization (Shalina). The result is an integrated, air-gapped defense platform proven under reproducible, rigorous evaluation."*